

Kanav Gupta

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RESEARCH INTERESTS

Secure Multi-Party Computation, Privacy Enhancing Technologies, Privacy-Preserving Machine Learning

EDUCATION

- **University of Maryland, College Park** College Park, MD, USA
PhD in Computer Science; Advisor: Prof. Jonathan Katz; GPA: 4.0 Aug 2023 - Aug 2028 (Expected)
- **Indian Institute of Technology Roorkee** Roorkee, India
B.Tech in Computer Science and Engineering; CGPA: 9.04/10 July 2017 - July 2021

PUBLICATIONS

- [1] Kanav Gupta, Nishanth Chandran, Divya Gupta, Jonathan Katz, and Rahul Sharma. “Shark: Actively Secure Inference using Function Secret Sharing”. In: *IEEE S&P* 2025.
- [2] Kanav Gupta, Neha Jawalkar, Ananta Mukherjee, Nishanth Chandran, Divya Gupta, Ashish Panwar, and Rahul Sharma. “SIGMA: Secure GPT Inference with Function Secret Sharing”. In: *PETS* 2024. Accepted for a talk at RWC’24 and TPMPC’24.
- [3] Neha Jawalkar, Kanav Gupta, Arkaprava Basu, Nishanth Chandran, Divya Gupta, and Rahul Sharma. “Orca: FSS-based Secure Training with GPUs”. In: *IEEE S&P*. 2024.
- [4] Kanav Gupta, Deepak Kumaraswamy, Nishanth Chandran, and Divya Gupta. “LLAMA: A Low Latency Math Library for Secure Inference”. In: *PETS*. 2022.

WORK EXPERIENCE

- **Google** New York City, NY
Software Engineering PhD Intern May 2025 - Aug 2025
 - Designed and implemented Garbled Circuits based protocol for secure multi-party differentially private training.
- **Microsoft Research India** Bengaluru, India
Research Intern May 2024 - August 2024
 - Designed protocols for secure inference of transformer models, secure against a malicious adversary.
- **Microsoft Research India** Bengaluru, India
Research Fellow July 2021 - July 2023
 - Worked on problems related to applications of secure multi-party computation to machine learning.
 - Published 3 research papers in secure inference and secure training.
 - Lead the development of *Sytorch*, a C++ framework that allows the developer to describe a machine learning model, similar to pytorch, and execute secure inference and training of the model using a variety of protocol options.
- **MaidSafe** Remote
Intern Feb 2021 - May 2021
 - Contributed to open-source development of the *self-encryption* toolkit - a component of Safe Network which splits data into chunks and encrypts each of these chunks with a key derived from subsequent chunk.
- **Simula UiB** Bergen, Norway
Research Assistant Sep 2020 - Dec 2020
 - Studied Shortest Vector Problem in Lattice-based Cryptography as a part of Bachelor’s project.
 - Introduced the notion of Obtuse Basis and showed that it is exponentially faster to solve SVP on such a basis.

REVIEW EXPERIENCE

Journals. ACM TOPS 2024

External Reviewer. ACM CCS 2023; Eurocrypt 2024,2026; Asiacrypt 2024; USENIX Security 2025,2026; Crypto 2025

HONORS AND AWARDS

- Awarded RSAC Security Scholar 2026
- Awarded Kulkarni Summer Research Fellowship 2024 (declined)
- Awarded funding to attend RWC 2024 and PETS 2024.
- Awarded Dean's fellowship at UMD
- Second place at Warpspeed web3 hackathon organized by Lightspeed Venture Partners.
- Bronze Medal in NSUCRYPTO Olympiad 2020.
- Winner of CSAW Embedded Security Challenge 2020 in India region as a part of team SDSLabs.
- Winner of GitHub Java CTF 2020.
- Winner of CSAW CTF 2019 in India region as a part of team InfoSecIITR.
- Winner of SecCon CTF 2019 hosted by Cisco India.
- Second Place in Warpspeed Hackathon 2022.
- Winner of UST Global d3code Hackathon 2019.
- Recipient of KVPY scholarship 2017 with All-India Rank 161.
- Secured All-India Rank 430 in JEE Advanced 2017 and All-India Rank 113 in JEE Main 2017.
- Rank 1 in Regional Mathematics Olympiad, KVS Region, 2015.

PROGRAMMING SKILLS

- **Skills:** Reverse Engineering, Low-Latency Programming, Emulation, web3
- **Languages:** C++, C, Python, Rust, Golang, OCaml, Solidity
- **Tools:** IDA Pro, Ghidra, Docker, XCode

TEACHING EXPERIENCE (TA)

- Governing Algorithms and Algorithmic Governance (with Prof. Gabriel Kaptchuk), UMD, Fall 2024
- Introduction to Quantum Computing (with Prof. Gorjan Alagic), UMD, Spring 2024
- Discrete Structures, UMD, Fall 2023
- Data Structures, IIT Roorkee, Spring 2020
- Discrete Structures, IIT Roorkee, Spring 2019

EXTRA CURRICULAR ACTIVITIES

- Currently serving as the organizer for UMD Crypto Reading Group. We meet every thursday!
- Served as a Joint Secretary of the technical group *SDSLabs*. I was responsible for organizing several open institute lectures on fundamental topics of computer science. Led several projects like Backdoor, Beast and Watchdog.
- Participated in and won numerous CTFs as a part of the team *SDSLabs*.
- Actively participated in open-source development of *DifferentialEquations.jl* - a toolchain to solve ordinary differential equations using numerical methods.